Mardi 12 mars 2026 – 17h08

Modèle Conceptuel de Données (MCD) – Version 1.0 du projet MyDigitrans.

Cette version est optimisée pour :

- Multi-agence
- Multi-devise
- Paiement mixte
- Gestion caisse
- Réservation / Paiement partiel / Billet final
- Module colis
-Module commission
- Historique simple sécurisé

1. ORGANISATION

1. Agency

- Name
- resgistrationNumber(*nullable*)
- taxNumber(nullable)
- Phone
- Email
- Address
- isActive
- Country
- City
- createdAt
- UpdatedAt(nullable)
- Slug
- Code
- Logo (nullable)
- DefaultCurrency(nullable)

2. Branch

- Name
- Agency
- City
- Country
- Status (*opened, closed, suspended*)
- Code
- Slug
- createdAt
- updatedAt(nullable)
- isActive

3. User

- firstname
- lastname
- phone
- email
- password
- userType(*platform,agency,broker*)
- sponsor (user)
- isActive
- lastLoginAt
- agency (nullable)
- brach(nullable)
- createdAt

4. Role (system)

- name
- code (*ROLE_SUPER_ADMIN, ROLE_CASHIER,…)*
- scope (platform,agency)
- description(nullable)
- isActive

5. UserRole

- user
- role
- agency(nullable)
- branch(nullable)
- assignedAt
- AssignedBy
- RevokedAt(nullable)
- revokedBy(nullable)

6. Permission (system)

- name (*lecture ticket*)
- code (ticket.view)
- description(*nullable, voir la liste des tickets vendu par exemple*)
- module(ticket,payment,agency,brach,trip etc.)
- createdAt
- createdBy
- scope(platform,agency)

7. rolePermission

- role
- permission
- createdAt
- branch(nullable)
- agency(nullable)
- assignedAt
- AssignedBy
- RevokedAt(nullable)
- revokedBy(nullable)

Role	Permission
Cashier	Ticket.create
Cashier	Ticket.view
Manager	Ticket.refubd

2. **TRANSPORT**

8. busLayout (System)

- name (*modèle 2-2 ; ; modèle 3-1 ; etc*)
- agency(*nullable*)
- rows
- leftColumns
- rightColumns
- middleColumns
- aisles(*string/Json position de couloirs ex : [" 2"] un couloir apres la deuxieme colonne*)
- specialPositions (*JSON*)
- hasBackExtraSeat(*bool*)
- createdAt

9. Bus

- agency
- busLayout
- code (nullable)
- name(nullable)
- plateNumber(unique)
- brand (toyota, mercedes etc)
- model (Tx, Maybach, etc)
- status(*active,maintenance,inactive*)
- year
- createdAt
- createdBy

10. Seat (automatique creation)

- bus
- seatLabel
- type(*driver, normal, vip, extra,wc*)
- row
- column
- createdAt

11. Route

- agency
- departureCity
- departureCountry
- arrivalCity
- arrivalCountry
- distance (km)
- duration
- createdAt
- createdBy
- isAvailable

12. Trip

- route
- bus
- departureDate
- departureTime
- arrivalDate
- arrivalTime
- status(*scheduled,boarding,departed,arrived,cancelled*)
- createdAt
- createdBy

13. TripSeat (*automatique after trip creation*)

- unique(trip,seat to avoid duplecation)
- trip
- seat
- seatStatus(*held, reserved, blocked, available*)
- reservation(*nullable*)
- heldUntil(*nullable*)

3 CLIENT

14. Customer

- firstName
- lastName
- phone
- email(nullable)
- agency
- createdBy
- createdAt

4 TARIFICATION

15. Currency

- code (USD, CDF) (unique)
- name (*dollar, Franc Congolais*)
- agency (*nullable*)
- createdAt
- createdBy

16. ExchangeRate(agency admin)

- baseCurrencyId
- targetCurrencyId
- rate
- createdAt
- agency(*nullable*)
- effectiveFrom

17. Fare

- fareType (*promo, standard,VIP*)
- route
- agency
- amount
- currency
- validFrom
- validUntil
- createdAt
- createdBy

5. VENTE BILLET

18. Reservation

- reference
- trip
- agency
- costumer
- status(*PENDING, CONFIRMED, CANCELLED,EXPIRED*)
- totalAmount
- currency
- createdAt
- createdBy
- expiresAt

➢ ***PENDING*** : Reserved but not paid
➢ ***TOTALAMOUNT :*** Je reserve par exemple 3 siege de 15$ le total sera 45$
➢ ***ExpiresAt :*** interaction avec tripseat, si la reservation expire, le seat devient available

19. Ticket

- Agency
- ticketNumber
- Reservation
- tripSeat
- travelerName
- travelerPhone
- price
- currency
- status (*active,cancelled,refunded,used*)
- createdAt
- createdBy

20. AgencyPayment

- type (*reservation, Shipment*)
- reference
- amount
- currency
- exchangeRateUsed
- reservation(nullable)
- shipment(nullable)
- paymentMethod (*cash, mobile_money, card,bank_transfert*)
- status(pending,failed,refunded,confirmed)
- createdBy (nullable)
- confirmedBy (nullable)
- confirmedAt
- createdAt
- agency
- branch

21. Refund

- payment
- agency
- amount
- currency
- reason
- status(pending,success,failed)
- createdBy
- createdAt

22. Discount

- referencedType (*reservation,shipment*)
- referencedId
- amount
- currency
- createdAt
- createdBy

23. TicketPrintLog

- ticketId
- printedBy
- printedAt
- reason (*original, duplicate*)
- printCount

24. TicketScanLog

- ticketId
- scannedBy
- scannedAt
- result (valid, already_used, invalid)

6. MODULE COLIS

25. Shipment

- reference
- tripId
- senderId (customerId)
- receiverId (customerId)
- description
- departureAgencyBranch
- arrivalAgencyBranch
- weight
- price
- currency
- agency
- branch
- status (Pending, in_transit,delivered, canceled)
- createdBy
- createdAt

7. **CAISSE**

26. CashRegister

- agency
- branch
- name (Guichet 1, etc)
- code
- currency
- status (active,inactive)
- createdAt
- createdBy

27. CashSession

- cashRegister
- openedBy
- openedAt
- closedBy
- closedAt
- status(open,closed)
- openingBalance
- closingBalance
- expectedBalance
- countedBalance
- difference
- notes (pour expliquer le manque par exemple)

28. Expense

- type (*subscription,other*)
- name(*abonnement mars 2026,transport agents,etc*)
- amount
- currency
- agency
- branch
- description
- status (pending, paid, failed)
- validFrom
- validUntil
- createdAt
- createdBy
- confirmedBy

29. FinancialOperation

- cashSessionId
- type (in, out)
- referenceType(agencyPayment,refund, expense)
- referenceId
- amount
- currencyId
- createdAt
- createdBy
- description

9. **SOUSCRIPTION**

30. Feature

- code (*online_booking,advanced_reports,shipment_module etc*)
- name (*Reservation en ligne, Gestion colis etc*)
- description
- status(*active, inactive*)
- createdAt
- createdBy

31. subscriptionPlan

- name (*basic,pro,trial etc*)
- price
- currency
- duration_days
- max_branches
- max_users
- max_buses
- status (*active, inactive*)
- createdAt
- createdBy

32. PlanFeature

- plan
- feature
- createdAt
- createdBy

33. subscription

- agency
- plan
- startAt
- endAt
- status (*active,expired,canceled,suspended*)
- createdAt
- createdBy

34. Invoice

- agency
- type (*subscription, platformFee, mixed, donc abonnemment, commision sur les ventes et les deux*)
- invoiceNumber
- amount
- currency
- status (*draft,issued,paid,cancelled,overdue*)
- issuedAt (*facture envoyée le*)
- dueDate(*date limite sinon le status change par ex : expired*)
- paidAt
- createdAt
- createdBy
- subtotal
- discountTotal
- total
- balanceDue (*reste à payer pour le cas de paiement partiel*)
- periodStart
- periodEnd

35. platformFeeRule

- sourceType(*reservation,shipment*)
- feeType (fixed,percent)
- value
- currency (*nullable pour le pourcentage*)
- agency (*nullable, si fixé, cette agence suivra cette rule*)
- createdAt
- createdBy

36. platformFee (*créer à la confirmation du paiement de la reservation ou colis***)**

- agency
- rule
- sourceType(*reservation,shipment*)
- sourceId
- amount (*si fixe rule.value, si pourcentage ticket.price*rule.value/100*)
- currency
- status (pending,invoiced,paid)
- invoice(*nullable*)
- createdAt

37. platformRefund

- invoice
- agency
- amount
- currency
- reason
- status(pending,success,failed)
- createdBy
- createdAt

38. PlatformDiscount (*definit les promotions***)**

- type (*fixed, percent*)
- name (*promo, bonus*)
- value
- currency(*nullable*)
- validForm
- validUntil
- maxUses (*limite d'utilisation*)
- usedCount (*nombre de fois utlisé)*
- status (*active,inactive,expired*)
- createdAt
- createdBy

39. InvoicePlatformDiscount

- invoice
- amount
- currency
- platformdiscount (*nullable*)

40. subscriptionPayment

- subscription
- invoice
- amount
- currency
- method (*manual, mobile money etc*)
- paidAt
- status (*pending,success,failed,refunded*)
- refrence
- confirmedBy (*nullable*)
- createdAt
- createdBy
- confirmedAt (*nullable*)

9. PLATFORME CAISSE

41. PlatformBranch

- name
- city
- adress
- status
- createdAt
- createdBy

42. platformCashRegister

- platformbranch
- name
- code
- currency
- status
- createdAt
- createdBy

43. platformCashSession

- platformCashRegister
- platformbranch
- openedBy
- openedAt
- closedBy
- closedAt
- openingBalance
- closedBalance
- countedBalance
- difference
- status (*closed, opened*)

44. PlatfomExpense

- name(*hebergement, loyer, transport agents,etc*)
- amount
- currency
- platformbranch
- description
- status (pending, paid, failed)
- validFrom
- validUntil
- createdAt
- createdBy
- confirmedBy

45. platformFinancialOperation

- platformCashSession
- operationType (*commission, subscription_payment, refund, expense, adjustment*)
- type (in, out)
- referenceType(invoice,refund,commissionPayment,platformExpense)
- referenceId
- amount
- currencyId
- createdAt
- createdBy
- description

10. COMMISSION

46. commissionRule

- name
- sourceType (*Ticket,Shipment,subscription*)
- level
- maxLevel
- percentage
- isActive
- createdAt
- createdBy

47. commission

- user
- sourceType
- sourceId
- level
- percentage
- amount
- currency
- createdAt
- status (*pending,approved,paid,cancelled*)
- approvedBy (*nullable*)
- approvedAt (*nullable*)

48. commissionPayment

- user
- amount
- currency
- method
- reference
- status (*pending, success, failed*)
- paidAt
- createdAt
- createdBy
- paidBy (*agency,platform*)
- paidById (*user*)

49. commissionPaymentItem

- commission (*unique*)
- commissionPayment
- amount
- createdAt

11 SÉCURITÉ & HISTORIQUE

50. AuditLog

- user
- action (*create, update, cancel, refund,login,delete,confirm_payment,cancel_booking*)
- entityName (resevation etc)
- entityId
- oldValues(*JSON ['status':'pending']*)
- newValue (*JSON ['status':success,*)
- createdAt
- ipAddress
- device_type(mobile,POS,Desktop)
- user_agent(Mozila 5.0 ; android chrome etc)

🎯 LOGIQUE DOCUMENTAIRE

Un seul Ticket en base.

Selon paymentStatus :

- unpaid → Bon de réservation
- partial → Reçu de paiement
- paid → Billet officiel

MyDigitrans – Architecture Système V1.0